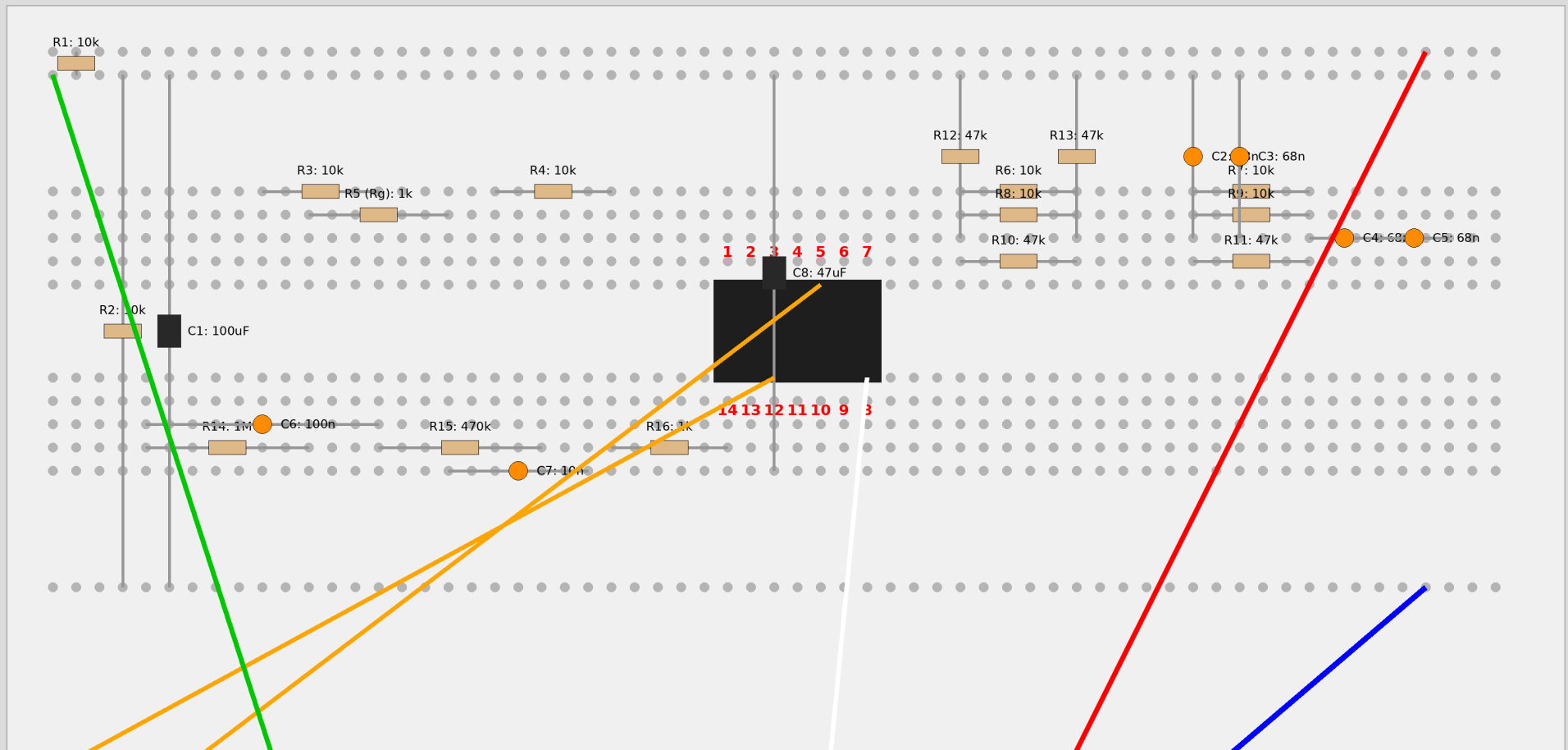


DIY EEG Circuit Guide (TL074 + Single 9V Battery)

EEG Circuit Schematic (Single TL074)

DIY EEG Circuit Guide (TL074 + Single 9V Battery)

AUDITED EEG LAYOUT - ALL 24+ COMPONENTS PRESENT



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DESIGN.md

EEG Circuit Design (Single TL074, Single 9V Battery)

Architecture

1. Virtual Ground (VGND): 4.5V reference using two 10k resistors and a 100uF capacitor.
2. Stage 1: Instrumentation Amplifier: 3-op-amp InAmp, Gain ~ 21 .
3. Stage 2: 50Hz Notch Filter: Passive Twin-T, tuned for 49.8Hz.
4. Stage 3: Gain + Bandpass Filter: Gain ~ 471 , Bandpass 3.4Hz - 34Hz.

Total Gain: $\sim 10,000$.

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ASSEMBLY.md

Breadboard Assembly Guide

Step 1: Power & VGND

- Red Rail: +9V. Blue Rail: 0V.
- VGND: Junction of two 10k resistors between +9V and 0V. Stabilize with 100uF to 0V.

Step 2: TL074 IC

- Side 1 (Bottom): Pins 1-7. Side 2 (Top): Pins 14-8.
- Pin 4 to +9V. Pin 11 to 0V.

Step 3: Stages

- Stage 1 (InAmp): Pins 12, 13, 14, 5, 6, 7, 2, 3, 1.
- Stage 2 (Notch): Pin 1 to Input, Output to Pin 10.
- Stage 3: Output at Pin 8.

Step 4: Connections

- Electrodes: Left side of board.
- Battery: Under the board.
- Oscilloscope: Bottom of board.

TESTING.md

Safety & Testing

Safety

1. Battery Only.
2. Oscilloscope Isolation.
3. Reference Grounding.

Testing

- Check VGND (4.5V).
- Short inputs to VGND, check output (Flat line).
- Capture Alpha waves (8-13Hz) with eyes closed.

CONNECTIONS.md

Detailed Connection List (Point-to-Point)

1. VGND Group

- R1 (10k): V+ to VGND
- R2 (10k): VGND to 0V
- C1 (100uF): VGND (+) to 0V (-)

2. IC Power

- Pin 4: V+
- Pin 11: 0V

3. InAmp (Stage 1)

- R5 (Rg, 1k): Pin 13 to Pin 6
- R3 (10k): Pin 14 to Pin 13
- R4 (10k): Pin 7 to Pin 6
- R6 (10k): Pin 14 to Pin 2
- R7 (10k): Pin 7 to Pin 3
- R8 (10k): Pin 3 to VGND
- R9 (10k): Pin 2 to Pin 1

4. Notch Filter (Stage 2)

- R10 (47k): Pin 1 to Node X
- R11 (47k): Node X to Node Y (Notch Out)
- C2/C3 (2x 68n parallel): Node X to VGND
- C4 (68n): Pin 1 to Node Z

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- C5 (68n): Node Z to Node Y
- R12/R13 (2x 47k parallel): Node Z to VGND

5. Output Stage (Stage 3)

- C6 (100n): Node Y to Pin 10
- R14 (1M): Pin 10 to VGND
- R15 (470k): Pin 8 to Pin 9
- C7 (10n): Pin 8 to Pin 9
- R16 (1k): Pin 9 to Node W
- C8 (47uF): Node W (+) to VGND